

```

ubuntu@ip-172-31-39-169: ~$ docker-compose version 1.29.2, build unknown
ubuntu@ip-172-31-39-169:~$ cd DevOps_Midterm/phase3

ubuntu@ip-172-31-39-169:~/DevOps_Midterm/phase3$ cd ..
ubuntu@ip-172-31-39-169:~/DevOps_Midterm$ docker build -f phase3/Dockerfile -t product-app:latest .
[*] Building 20.7s (15/15) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 1.49kB                             0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine  2.6s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [builder 1/4] FROM docker.io/library/node:20-alpine@sha256:b88333c42c23fbd91596ebd7fd10de239cedab9617de04142d  3.7s
=> => resolve docker.io/library/node:20-alpine@sha256:b88333c42c23fbd91596ebd7fd10de239cedab9617de04142dde7315e3  0.0s
=> => sha256:9d10d4687fae613c052431ea3221ccda835c3262df3564310345ffe173146d9 445B / 445B  0.5s
=> => sha256:7ad115895a6aadccf7f98a05c33873c04df2933ba557c5bcdbedd3de612803da 1.26MB / 1.26MB  0.9s
=> => sha256:589902ba0eae121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153 3.86MB / 3.86MB  1.1s
=> => sha256:ed2fdcee5269a06b08b77c69867176443364a72a9626e9aa2e66ab1ebda35a7 43.22MB / 43.22MB  1.9s
=> => extracting sha256:589902ba0eae121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153  0.4s
=> => extracting sha256:ed2fdcee5269a06b08b77c69867176443364a72a9626e9aa2e66ab1ebda35a7  1.5s
=> => extracting sha256:7ad115895a6aadccf7f98a05c33873c04df2933ba557c5bcdbedd3de612803da  0.1s
=> => extracting sha256:9d10d4687fae613c052431ea3221ccda835c3262df3564310345ffe173146d9  0.0s
=> [internal] load build context                                  2.0s
=> => transferring context: 19.38MB                             1.9s
=> [builder 2/4] WORKDIR /app                                    0.1s
=> [stage-1 2/7] RUN apk add --no-cache dumb-init                1.8s
=> [builder 3/4] COPY phase1/app/package*.json ./                0.1s
=> [builder 4/4] RUN npm ci --only=production --omit=dev          5.3s
=> [stage-1 3/7] RUN addgroup -g 1001 -S nodejs && adduser -S nodejs -u 1001  0.4s
=> [stage-1 4/7] WORKDIR /app                                    0.1s
=> [stage-1 5/7] COPY --from=builder --chown=nodejs:nodejs /app/node_modules ./node_modules  0.5s
=> [stage-1 6/7] COPY --chown=nodejs:nodejs phase1/app/ ./        4.1s
=> [stage-1 7/7] RUN mkdir -p /app/public/uploads && chown -R nodejs:nodejs /app/public/uploads  0.3s
=> exporting to image                                           3.2s
=> => exporting layers                                          1.7s
=> => exporting manifest sha256:a5d7a47d1290cb42a110d3777d0829cb15e2ec4fae96e361de7c199f3cd90a2  0.0s
=> => exporting config sha256:a769be41a0e03a6a5a4c20b2bf1409e811446159ce93e3d03957c679807326  0.0s
=> => exporting attestation manifest sha256:668d65fa0493e819eaac59f87ca965c2039c0f582b5432fb0f09c9f5d2bd5c105  0.0s
=> => exporting manifest list sha256:3ce7b8c4232053dc524000bad32dc6ba51766b6f59f0507f5ac51e613493c91d  0.0s
=> => naming to docker.io/library/product-app:latest            0.0s
=> => unpacking to docker.io/library/product-app:latest         1.4s
ubuntu@ip-172-31-39-169:~/DevOps_Midterm$ docker run -p 3000:3000 product-app:latest

```

Image 1: Build docker image

```

ubuntu@ip-172-31-39-169:~/DevOps_Midterm$ docker tag product-app:latest huytran905/product-app:latest
ubuntu@ip-172-31-39-169:~/DevOps_Midterm$ docker push huytran905/product-app:latest
The push refers to repository [docker.io/huytran905/product-app]
4f4fb700ef54: Layer already exists
9d10d4687fae: Layer already exists
589902ba0eae: Layer already exists
7ad115895a6a: Layer already exists
6fed013d7da0: Pushed
29dafcd1a254: Pushed
ed2fdcee5269: Layer already exists
53c992f82e6a: Pushed
b9525104d40a: Pushed
d99f75fec870: Pushed
7b8727b41548: Pushed
latest: digest: sha256:3ce7b8c4232053dc524000bad32dc6ba51766b6f59f0507f5ac51e613493c91d size: 856
ubuntu@ip-172-31-39-169:~/DevOps_Midterm$ cd phase3

```

Image 2: Push image to docker hub

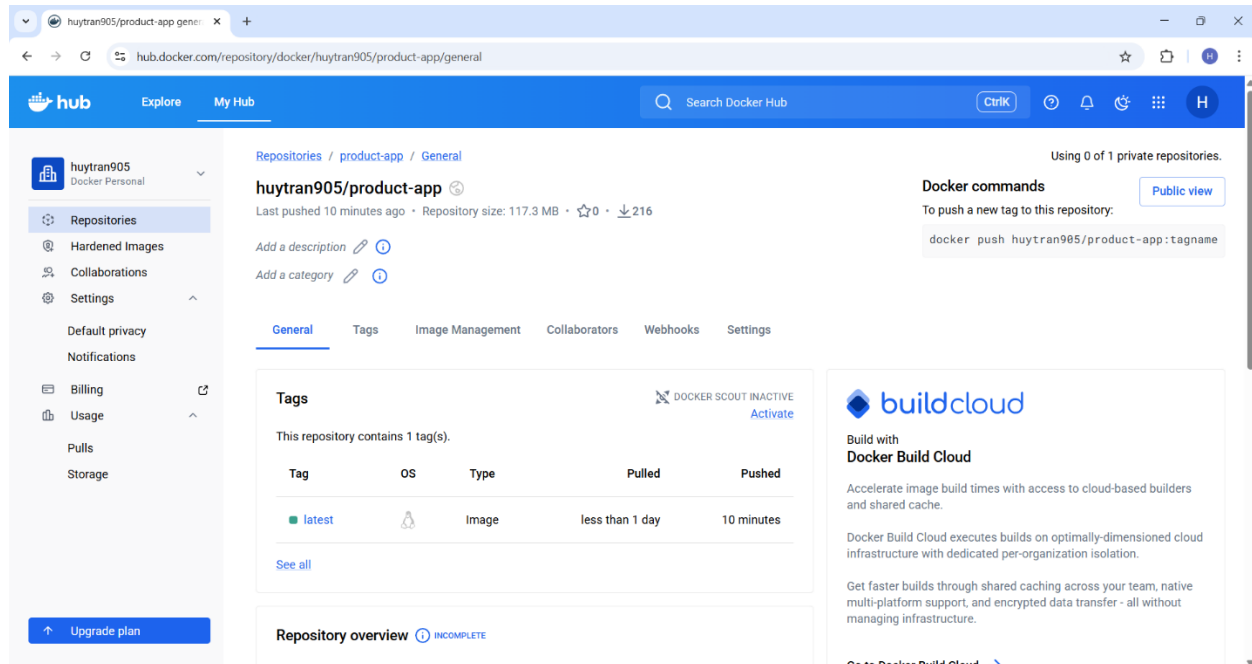


Image 3: Docker Hub displays the tag latest

STEP 4: Pulling Docker Image

```
Pulling image: huytran905/product-app:latest
latest: Pulling from huytran905/product-app
Digest: sha256:347b5dbcbf0292e86040fe08dab1e533e74e01ab4ba3b941461802f4ab7011b7
Status: Image is up to date for huytran905/product-app:latest
docker.io/huytran905/product-app:latest
✓ Image pulled successfully
```

STEP 5: Starting Services

```
Starting services with Docker Compose...
WARN[0000] /home/ubuntu/DevOps_Midterm/phase3/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 6/6
✓Network product-app-network Created 0.1ss
✓Volume product_mongodb_config Created 0.0ss
✓Volume product_uploads_data Created 0.0ss
✓Volume product_mongodb_data Created 0.0ss
✓Container product-mongodb Healthy 12.4s
✓Container product-web Started 12.6s

Waiting for services to be healthy...
```

Image 4: Docker containers started using Docker Compose

```
ubuntu@ip-172-31-28-22:~/DevOps_Midterm/phase3$ docker compose ps
WARN[0000] /home/ubuntu/DevOps_Midterm/phase3/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS                PORTS
product-mongodb     mongo:6.0            "docker-entrypoint.s..."  mongodb    8 minutes ago  Up 8 minutes (healthy)  27017/tcp
product-web         huytran905/product-app:latest "dumb-init -- node m..."  web        8 minutes ago  Up 8 minutes (healthy)  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
ubuntu@ip-172-31-28-22:~/DevOps_Midterm/phase3$
```

Image 5: List image using Docker Compose

```
ubuntu@ip-172-31-28-22:~/DevOps_Midterm/phase3$ docker volume ls
DRIVER      VOLUME NAME
local       product_mongodb_config
local       product_mongodb_data
local       product_uploads_data
ubuntu@ip-172-31-28-22:~/DevOps_Midterm/phase3$ |
```

Image 6: List volume using Docker Compose

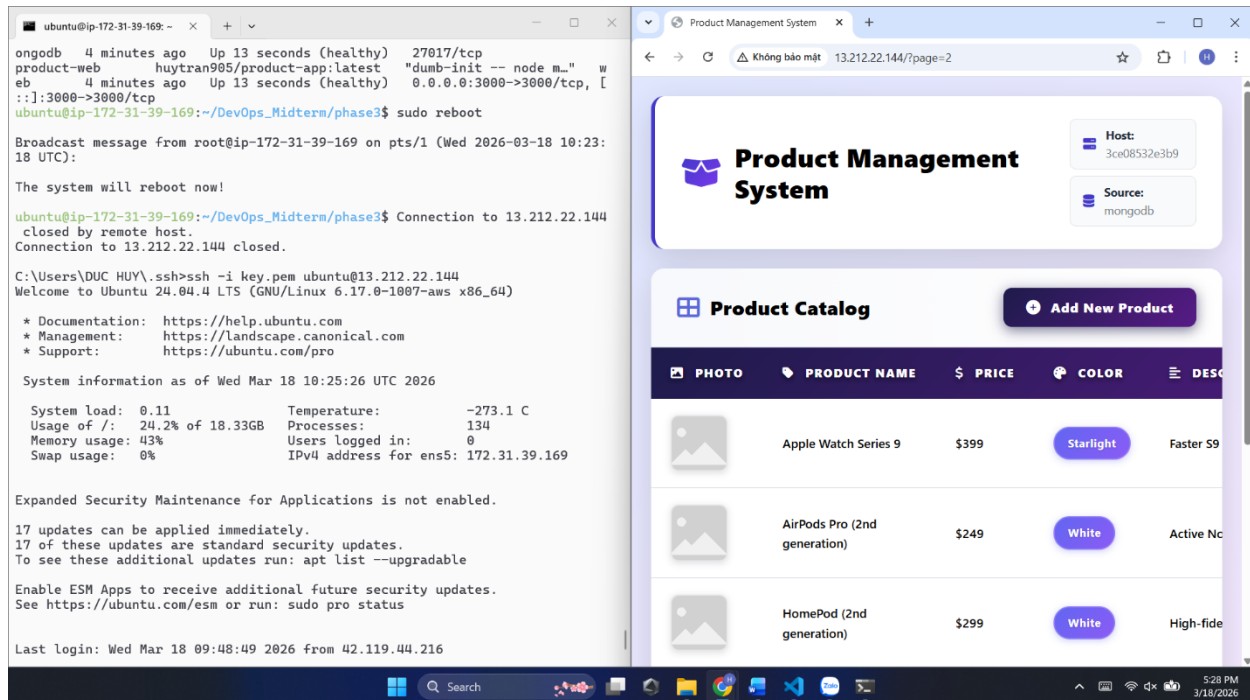


Image 7: The old data remains after the reboot